

Submission LINALG1-PS2-SUB02

Question LINALG1-PS2-Q01

work / setup:

1. $\det=6(5)-3(1)=27$

2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$

crossed-out arithmetic: [smudged]

FINAL (maybe): $x=1$, $y=2$ plus 1

Question LINALG1-PS2-Q02

work / setup:

1. $\det(A)=4 \cdot 4 - 2 \cdot 2 = 12$

crossed-out arithmetic: [smudged]

FINAL (maybe): $\det=12$; invertible plus 1

Question LINALG1-PS2-Q03

work / setup:

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$

2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear plus 1

Question LINALG1-PS2-Q04

work / setup:

1. Set $c \blacksquare (1, 0) + c \blacksquare (0, 1) = (0, 0)$.

crossed-out arithmetic: [smudged]

FINAL (maybe): only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent plus 1

Question LINALG1-PS2-Q05

work / setup:

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.

crossed-out arithmetic: [smudged]

FINAL (maybe): the vectors mapped to zero; exactly the solution space of $Ax=0$ plus 1

Question LINALG1-PS2-Q06

work / setup:

1. $\det=6(3)-2(3)=12$

2. Eliminate y , yielding $x=1$; substitute $\Rightarrow y=2$

crossed-out arithmetic: [smudged]

FINAL (maybe): $x=1$, $y=2$ plus 1

Question LINALG1-PS2-Q07

work / setup:

1. $\det(A)=6 \cdot 2 - 2 \cdot 2 = 8$

crossed-out arithmetic: [smudged]

FINAL (maybe): $\det=8$; invertible plus 1

Question LINALG1-PS2-Q08

work / setup:

1. $T(1, -1) = (1 - 2, 3 + 1) = (-1, 4)$

2. Matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear plus 1